

A Model of the Attention Economy: Data, Platforms, and Discovery[†]

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This paper develops a dynamic model of the attention economy featuring two interlinked markets: a product market, where consumers have limited awareness of firms, and an attention market, where platforms attract attention by investing in free services (e.g., social media, streaming, podcasts) and sell targeted ads using data on consumer preferences. The model provides a tractable framework in which platform investment, ad-driven discovery, product demand, and ad revenue are jointly determined. The effects of data and interoperability policies can reverse over time, platform quality may be inefficiently high or low, and the rise of data-rich digital advertising can coincide with a declining ad revenue-to-GDP ratio. (JEL D43, D44, D58, D83, L86, M37)

“...a wealth of information creates a poverty of attention and a need to allocate that attention efficiently...”

—Herbert Simon¹

Simon’s insight captures today’s digital economy, where social media, news, and entertainment compete intensely for attention. Individuals now spend over two and a half hours per day on digital media (Evans 2020). This attention underlies the Internet’s core business model: Platforms (any website or app displaying ads) collectively earn hundreds of billions of dollars each year from data-driven targeted advertising (Silk and Berndt 2021).

How do these platforms mediate the allocation of attention between consumers and firms, and how does this allocation shape demand, competition, and welfare over time?

A core challenge is that the attention economy involves a complex interaction between two markets. In the product market, consumers can only purchase from

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¹ Simon, Herbert. 1971. “Designing Organizations for an Information-Rich World,” In *Computers, Communications, and the Public Interest*, edited by Martin Greenberger, 40–41. Johns Hopkins Press.

firms they are aware of. In the attention market, platforms provide consumers with services (e.g., social media, streaming, podcasts) in exchange for attention, and sell targeted ads that determine which firms consumers discover in the product market. Ads are allocated through auctions in which firms bid on individual impressions using platform data on consumers' preferences. Because services are often free, platforms compete through investment in service quality.

Many key issues—the welfare tradeoffs between paid products and free platform services, and the cross-market effects of policy—cannot be studied without modeling these markets jointly. This paper develops a tractable framework that does so: Platforms' investment in quality drives attention, attention drives ad-funded discovery, discovery shapes product-market competition, and product-market competition in turn shapes the ad revenue that funds platform investment. To illustrate its value, I use the framework to study the dynamic effects of data and interoperability policies, the welfare properties of equilibrium, and how asymmetric data shapes competition between platforms such as traditional and digital media.

To keep the model tractable, I impose simplifying assumptions. In the baseline, platforms are monopolistically competitive: They share business models with real-world players such as Pinterest, Spotify, Meta, and YouTube, but do not interact strategically. The paper's focus is, thus, on market operation, *not* market concentration; a duopoly extension incorporates strategic investment under additional structure. Constant elasticity of substitution (CES) preferences over products and platform services further simplify matters: On the product side, prices are fixed and firms *optimally* do not personalize them. In return, the framework treats data flexibly² and allows multihoming by both firms and consumers, with a key parameter—the elasticity of attention to platform quality—governing platform competition and market power.

The paper yields three sets of findings, each pointing to the importance of cross-market spillovers for economic analysis. First, the effects of data and interoperability policies can be counterintuitive and reverse over time, as interventions trade off product-market efficiency against platform quality. Second, platform investment can be inefficiently high or low, and I derive a sufficient statistic that determines the sign of the distortion. Third, though platforms can profit from access to richer data than their rivals, *total* ad revenue can *fall*.

I now describe these findings in greater detail. I begin with numerical policy experiments highlighting mechanisms related to (i) data reforms and (ii) pro-competitive policies, such as interoperability mandates, that raise the elasticity of attention to platform quality and ad load.³ Although these policies originate in the attention market, endogenous product-market responses can overturn their initial effects.

In one experiment, giving platforms access to more informative data improves targeting and raises ad revenue in the short run. Over time, however, firms anticipate that any consumer they reach will have also been reached by better-matched competitors. This reduces a firm's willingness to pay for ads, causing long-run ad revenue to fall. Product consumption rises, but platforms invest less in quality.

²Data is modeled as signals about consumer preferences, without parametric restrictions on signal structure.

³Such policies are intended to make platform markets more contestable by giving consumers a more "meaningful choice" among platforms. For example, a content interoperability mandate requires platforms to allow users to share content or posts across sites. As content becomes more overlapping, attention becomes more sensitive to platform-specific features such as the interface or ad load.

In a second experiment, an interoperability mandate raises the elasticity of attention. Because consumers view ads as nuisances that reduce effective platform quality, platforms initially reduce ad load, depressing ad revenue and, in turn, investment. Lower ad load, however, weakens product discovery. Over time, firms anticipate fewer competitors for any consumer they reach, raising their willingness to pay for ads. In the long run, ad revenue and investment therefore rise. Product consumption falls even as platform quality improves.

I discuss conditions under which these patterns appear more likely to arise.

Next, I compare equilibrium to the planner's allocation. Two forces drive inefficiency. First, platforms compete for ad revenue, which creates business-stealing externalities. Second, because platforms charge users zero prices, they do not internalize the consumer surplus their services directly generate. These forces can push ad display rates and platform investment either above or below efficient levels. When consumers have Cobb-Douglas utility over CES aggregates of products and platform services, I derive a simple sufficient statistic⁴ that determines the sign of the investment distortion.

The policy experiments described earlier are intentionally stylized to isolate cross-market spillovers and dynamic effects, frequent policy focal points. In practice, interoperability policies often target large platforms, where strategic interaction may matter. Motivated by this, I next consider a duopoly model that preserves most of the baseline structure but takes ad display rates as exogenous, allowing platforms to interact strategically through investment. I develop a numerical algorithm to compute equilibrium and find that the mechanisms in the baseline policy experiments remain robust.

Finally, I extend the baseline model to allow two platform groups—e.g., digital and traditional media—with asymmetric data. The extension shows how data differences shape relative quality and market shares, and can generate a pattern consistent with a puzzling empirical fact: Despite the rise of highly profitable digital platforms, advertising's share of GDP fell in the period from 2000 to 2018 (Silk and Berndt 2021). One might expect the presence of data-rich platforms to raise ad revenue: Better matching creates surplus for firms, and platforms should be able to extract a share of the surplus gain. In a numerical example, however, digital platforms displace traditional media, earning a higher share of revenue due to their more informative data—yet total ad revenue declines.⁵ The result hinges on product-market competition: Better data improves the matching of *all* firms with consumers, so individual advantages erode and profit gains are offset, limiting the revenue platforms can extract.⁶

Beyond these main findings, the Supplemental Appendix develops further extensions (network effects, platform heterogeneity, and entry), which illustrate the model's potential as a foundation for future work.

The paper proceeds as follows. Section I provides background on the attention economy and related work. Section II introduces a simple product market that serves as a building block for the analysis. Section III integrates the attention market with the product market to develop the baseline model. Section IV characterizes

⁴The sufficient statistic depends on ad revenue, income, the product markup, platform substitutability, and the Cobb-Douglas weight.

⁵In the model, entry without data asymmetry cannot explain this: In fact, total ad revenue is *invariant* to the mass of platforms.

⁶In fact, with fixed prices and binding budget constraints, firms' profits gross of ad costs are constant.

equilibrium. Section V presents policy experiments, Section VI a welfare analysis, Section VII a duopoly extension, and Section VIII a model with asymmetric data. Section IX summarizes further extensions. Section X concludes.

I. Background

The term *attention economy* is now widely used to describe markets where attention is a scarce resource and a basis for economic exchange. It is often invoked in the context of digital advertising, linking it to issues of data use, platform design, and competition. These markets are inherently multisided: Platforms intermediate between consumers and advertisers while shaping the allocation of attention through the quality and features of their services. Competition is shaped not only by prices—many services are offered at zero monetary cost to consumers—but also by nonprice dimensions such as quality, data, and interoperability. These features pose distinctive challenges for economic analysis and policy, particularly when interventions on one side of the market have unintended consequences on others (OECD 2018, 2021, 2022).

This paper provides a transparent benchmark for analyzing these forces. The monopolistic competition baseline clarifies mechanisms that arise from the structure of the attention economy rather than strategic interaction, and lends itself to extensions. Section VII turns to a duopoly setting to study strategic interaction along the investment margin.

Prior work on digital advertising platforms (e.g., Bergemann and Bonatti 2023; Ambrus, Calvano, and Reisinger 2016; Prat and Valletti 2022; Galperti and Perego 2022) typically models large platforms in static settings and abstracts from either a microfounded product market or content provision.⁷ Similarly, Kirpalani and Philippon (2021) study matching in a large online marketplace but without content provision. Relative to this literature, my framework models the *attention economy*—bringing platform investment, ad-driven discovery, and product-market competition into a single system whose joint dynamics shape equilibrium outcomes.

A line of research in traditional advertising endogenizes free platform content provision, most notably Anderson and Coate (2005). Whereas Anderson and Coate (2005) model investment in entry of new platforms, I study investment in platform quality. Both models predict potential over- or under-investment due to business-stealing externalities and platforms' inability to charge consumers. However, in their setting consumers derive no surplus from advertising and there is no role for data or targeting.

In digital advertising, ads are commonly sold via auctions at the *individual* level in *real time* as consumers engage with platforms' services. Advertisers rely on platform data to tailor bids. Recent work investigates the value of information in auctions (e.g., Board 2009; Hummel and McAfee 2016; Bergemann et al. 2022; De Cornière and De Nijs 2016), but with few exceptions, abstracts from the product market. A central result is that more informative data typically raises revenue (Board

⁷ Ambrus, Calvano, and Reisinger (2016) and Prat and Valletti (2022) model the product market in reduced form. Bergemann and Bonatti (2024) and Galperti and Perego (2022) model a single platform and abstract from content and attention.

2009; Hummel and McAfee 2016). In my setting, richer data can reduce ad revenue because of spillover effects on advertisers' profits in the product market.

While the role of data in the economy has been examined in general equilibrium settings before (Jones and Tonetti 2020; Farboodi and Veldkamp 2026), it has rarely been studied in connection with advertising.⁸ A nascent literature incorporates advertising into macroeconomic models, including contemporaneous work by Cavenaile, Celik, Perla, and Roldan-Blanco (2026) and Rachel (2024), but without modeling data. Most closely, Greenwood, Ma, and Yorukoglu (2025) study advertising with free media goods in a static, perfectly competitive setting, where targeting distinguishes between college and noncollege consumers.

II. Simple Model of a Product Market

The baseline model has two components: a product market and an attention market. This section analyzes the product market in isolation—a simple monopolistically competitive environment isomorphic to the one in Melitz (2003). Although static, it is embedded in continuous time to serve as the backbone for the dynamic framework developed in the next section, where it is integrated with the attention market.

Time t takes values in $[0, \infty)$. A continuum of consumers $i \in \mathbb{I}$ of unit mass have heterogeneous tastes or “values” $\{v_{ij}\}$ for a continuum of products $j \in \mathbb{J}$ of mass J . Each product is produced by a distinct firm, also indexed by j for simplicity.

Consumer i has CES preferences for products,

$$C_{it} \equiv \left[\int_{\mathbb{J}} v_{ij}^{\frac{1}{\sigma}} c_{ijt}^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ is the elasticity of substitution across products.⁹ Values $\{v_{ij}\}$ are drawn independently across consumers and firms from a distribution F supported on $[0, \infty)$ with finite mean and remain constant over time.

At each time t , consumer i has income I and allocates it across products to maximize C_{it} taking firms' prices $\{p_{jt}\}$ as given. Firms set prices to maximize expected profits given a common constant marginal cost normalized to one (the numeraire). I make *no* assumptions about what firms know about consumers' values and could allow firms to personalize prices, though it turns out that firms would *optimally* choose not to do so in equilibrium.

The key friction in this otherwise standard model is that each consumer i is aware of only a subset $\Omega_{it} \subseteq \mathbb{J}$ and can consume only from this set. All that will matter for equilibrium is the mass M_t of firms in Ω_{it} and the consumer's average value $\mu_{\Omega_{it}} \equiv \frac{1}{M_t} \int_{\Omega_{it}} v_{ij} dj$ for those firms. I assume both objects are common across consumers.

PROPOSITION 1: *In the unique equilibrium of the product market:*

1. *Firm j sets price,*

$$(1) \quad p_{jt} = \frac{\sigma}{\sigma - 1} \cdot$$

⁸ These papers analyze data as an input into firm production. See also Demirer, Hernández, Li, and Peng (2024).

⁹ Values enter with exponent $1/\sigma$ as a normalization for notational convenience.

2. Firm j 's flow profit from selling to consumer i is $\pi_{\mathbb{J}t}v_{ij}$, where

$$(2) \quad \pi_{\mathbb{J}t} = \frac{I}{\sigma M_t \mu_{\Omega_t}}.$$

3. Consumer i 's aggregate consumption is

$$(3) \quad C_{it} = \frac{\sigma - 1}{\sigma} I (M_t \mu_{\Omega_t})^{\frac{1}{\sigma-1}}.$$

Intuition is provided in the Appendix. As seen from (1), CES preferences imply a constant markup independent of consideration sets or information firms hold about consumers' values.¹⁰ This property simplifies integration with the attention market because prices are fixed. Moreover, flow profits (2) and aggregate consumption (3) depend only on a single sufficient statistic: the cumulative value $M_t \mu_{\Omega_t}$ of firms in consideration sets. The attention market will endogenize the evolution of M_t and μ_{Ω_t} .

III. Baseline Model of the Attention Economy

This section integrates the product market with the attention market to form the baseline model of the attention economy. In what follows, I treat the microfounded relationships (1)–(3) as primitives.

The attention market consists of a continuum of platforms $k \in \mathbb{K}$ of mass K . Like product firms, platforms are monopolistically competitive: Each offers a distinct service (e.g., streaming, social media, podcasts) and thus has market power but is atomistic. Services are free but require attention to consume. Platforms provide these services to attract attention for ad display and invest to maintain their quality.

If a consumer allocates attention x_{ikt} to platform k , she views ads at Poisson intensity $a_{kt}x_{ikt}$, where a_{kt} is the platform's chosen *ad display rate*. Each ad opportunity is sold via a second-price auction the instant it arises. Due to latency, only N firms, drawn uniformly at random from $\Omega_{it}^c \equiv \mathbb{J} \setminus \Omega_{it}$ (outside the consideration set), submit bids within the instant.¹¹ The highest bidder wins and pays the second-highest bid. The winning firm enters Ω_{it} at the end of the instant and exits (i.e., is “forgotten”) at an independent exponential time with rate λ_f , ensuring that search frictions persist in the long run.

Firms bid based on signals of consumers' values for their products. These signals, interpreted as *platform-supplied data*, induce posterior expectations $\{\hat{v}_{ij}\}$ of consumers' values $\{v_{ij}\}$. Expectations are drawn independently from a continuous c.d.f. G , are constant over time, and do not depend on platform identity, since all platforms provide the same data.¹² Section VIII relaxes this last assumption.

I begin by describing how consideration sets evolve, taking attention, ad display rates, and bidding strategies as given. I then characterize agents' decision problems. Because consumers allocate attention symmetrically in equilibrium, I suppress the index i on x_{ikt} when convenient.

¹⁰The optimality of a constant markup is a feature of CES demand. The Appendix explains why it still holds here with heterogeneous values.

¹¹Firms only target consumers they do not already serve. A firm does not benefit from displaying an ad to a consumer whose consideration set the firm is already in.

¹²By Blackwell (1953), G is a mean-preserving contraction of the prior F over values but is otherwise unrestricted.

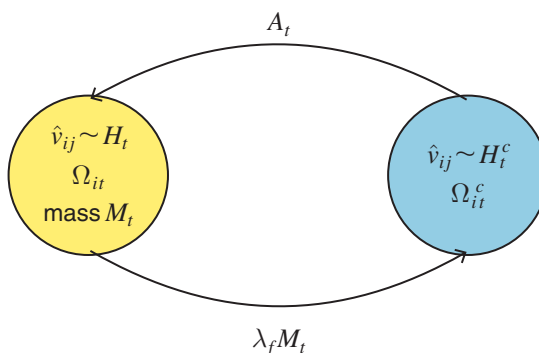


FIGURE 1. EVOLUTION OF CONSIDERATION SETS

Evolution of Consideration Sets.—The key state variables are the mass M_t and c.d.f. H_t of *expected* values for firms in Ω_{it} , both common across consumers. By the law of large numbers,¹³ the mean μ_{H_t} coincides with the mean μ_{Ω_t} of *actual* values in Ω_{it} . Since $M_t \mu_{\Omega_t}$ is a sufficient statistic for firms' profits and consumption, we do not need to track the true value distribution.

Figure 1 summarizes the evolution of the state variables. The c.d.f. H_t^c corresponds to expected values in Ω_{it}^c and is pinned down given H_t via an accounting identity:¹⁴

$$(4) \quad M_t H_t(\hat{v}) + (J - M_t) H_t^c(\hat{v}) = JG(\hat{v}), \quad \hat{v} \in [0, \infty).$$

The rate of inflow into Ω_{it} is

$$(5) \quad A_t \equiv \int_{\mathbb{K}} a_{kt} x_{kt} dk.$$

In equilibrium, the winning firm has the highest expected value, so the c.d.f. of inflowing firms is $(H_t^c)^N$. Because firms exit at random, outflow occurs at rate $\lambda_f M_t$ with CDF H_t . Thus,

$$(6) \quad \frac{d}{dt}[M_t H_t(\hat{v})] = A_t H_t^c(\hat{v})^N - \lambda_f M_t H_t(\hat{v}), \quad \hat{v} \in [0, \infty).$$

Taking $\hat{v} \rightarrow \infty$ yields¹⁵

$$(7) \quad \dot{M}_t = A_t - \lambda_f M_t.$$

Together, (4) and (6) fully characterize the dynamics of the key state variables.

¹³Throughout, I assume the law of large numbers. This can be formalized using methods from Duffie, Qiao, and Sun (2025).

¹⁴The mass of firms with expected values less than $\hat{v}$ in both Ω_{it} and Ω_{it}^c must coincide with that of the whole economy.

¹⁵Equations (6) and (7) apply whenever $M_t < J$ and I impose parameter conditions in Section IV so this is always so. The conditions for interchanging the derivative and limit hold here (but (7) can alternatively be derived from aggregate flow balance).

Firms' Bidding Problem.—Anticipating M_t and H_t , firms choose bids to maximize discounted profits. The formal problem is stated in Supplemental Appendix B; here I describe it recursively.

Because auctions are second price, firm j 's optimal bid for a consumer i with $\hat{v}_{ij} = \hat{v}$ equals the *gain* in continuation value from entering Ω_{it} :

$$(8) \quad B_t(\hat{v}) = V_t^{\text{In}}(\hat{v}) - V_t^{\text{Out}}(\hat{v}).$$

V_t^{In} and V_t^{Out} are the present values of profits from selling to consumer i when the firm is currently inside and outside Ω_{it} , respectively. While outside, firm j enters Ω_{it} with Poisson intensity $\lambda_{et}(\hat{v})$, characterized shortly. Given discount rate $\rho > 0$, V_t^{In} and V_t^{Out} satisfy the Hamilton-Jacobi-Bellman (HJB) equations:

$$(9) \quad \dot{V}_t^{\text{In}}(\hat{v}) = \rho V_t^{\text{In}}(\hat{v}) - \underbrace{\lambda_f [V_t^{\text{Out}}(\hat{v}) - V_t^{\text{In}}(\hat{v})]}_{\text{loss from exit}} - \underbrace{\pi_{jt} \hat{v}}_{\text{expected flow profit}}$$

$$(10) \quad \dot{V}_t^{\text{Out}}(\hat{v}) = \rho V_t^{\text{Out}}(\hat{v}) - \lambda_{et}(\hat{v}) \left(\underbrace{V_t^{\text{In}}(\hat{v}) - V_t^{\text{Out}}(\hat{v})}_{\text{gain from entry}} - \underbrace{\mathbb{E}[B_t^{(1)} | B_t(\hat{v}) > B_t^{(1)}]}_{\text{expected ad payment}} \right).$$

Here, $B_t^{(1)}$ is the highest bid among the $N - 1$ other bidders. If B_t is increasing, then $B_t^{(1)} \leq B_t(\hat{v}^{(1)})$ where $\hat{v}^{(1)}$ is the maximum of $N - 1$ draws from H_t^c .

In that case, accounting implies¹⁶

$$(11) \quad \lambda_{et}(\hat{v}) = \underbrace{\frac{NA_t}{J - M_t}}_{\text{auction entry intensity}} \underbrace{H_t^c(\hat{v})^{N-1}}_{\text{win probability}}.$$

Platforms' Ad and Investment Decisions.—Platforms choose ad display and investment to maximize discounted profits. Given B_t , the expected ad price is $\pi_{\mathbb{K}t} = \mathbb{E}[B_t(\hat{v}_t^{(2)})]$, where $\hat{v}_t^{(2)}$ is the second-highest of N independent draws from H_t^c . Anticipating $\pi_{\mathbb{K}t}$, platform k solves

$$(12) \quad \max_{\{a_{kt}\}, \{\ell_{kt}\}} \int_0^\infty e^{-\rho t} (\pi_{\mathbb{K}t} a_{kt} x_{kt}(a_{kt}, q_{kt}) - \ell_{kt}) dt$$

subject to the law of motion for quality:

$$(13) \quad \dot{q}_{kt} = \ell_{kt}^\varphi - \delta q_{kt}, \quad q_{k0} = q_0,$$

where $\varphi < 1$ ensures decreasing returns and $\delta > 0$ is the depreciation rate. The notation $x_{kt}(a_{kt}, q_{kt})$ indicates that the platform internalizes how its choices affect consumer attention.

Consumers' Attention Choices.—Consumers derive utility from both products and platform services,

$$U_i \equiv \int_0^\infty e^{-\rho t} u(C_{it}, X_{it}) dt,$$

¹⁶Each ad display triggers an auction with N invited bidders, so invitations are sent at total rate NA_t . The first term in (11) divides this rate by the mass of firms outside Ω_{it} .

where $u(\cdot)$ is increasing in both arguments. Platform services enter utility through a CES aggregator,

$$X_{it} \equiv \left[\int_{\mathbb{K}} (\nu(a_{kt}) q_{kt} x_{ikt})^{\frac{\epsilon-1}{\epsilon}} dk \right]^{\frac{\epsilon}{\epsilon-1}},$$

where $\nu(\cdot)$ is positive and decreasing so ads are nuisances, and $\epsilon > 1$ is the elasticity of substitution across platforms.

At each t , consumer i has one unit of attention to allocate across platforms. For simplicity, she does *not* internalize how attention affects $\{C_{is}\}_{s \geq t}$ through the evolution of her consideration set. In equilibrium, this myopia is without loss of optimality: Because all platforms set the same ad display rate, how the consumer divides attention among them is inconsequential. She thus solves

$$(14) \quad \max_{\{x_{ikt}\}} X_{it} \quad \text{s.t.} \quad \int_{\mathbb{K}} x_{ikt} dk \leq 1.$$

Attention is therefore allocated independently of the utility function u .

Equilibrium Concept.—

DEFINITION 1: *Taking the product market relationships (1)–(3) as primitives, an equilibrium of the economy with initial condition (M_0, H_0, q_0) is a collection of processes,*

$$\left\{ \{M_t\}, \{H_t\}, \{q_{kt}\}, \{B_t(\cdot)\}, \{a_{kt}\}, \{\ell_{kt}\}, \{x_{ikt}(\cdot)\} \right\},$$

such that (i) firms, platforms, and consumers solve their respective problems; and (ii) $\{M_t\}$, $\{H_t\}$, and $\{q_{kt}\}$ satisfy their laws of motion.

A. Modeling Remarks

The baseline model adopts simplifying assumptions to provide a transparent benchmark. Several are relaxed in later sections or online appendices, as summarized in the introduction. Beyond these, one can show that the results extend to any standard auction format by revenue equivalence. Moreover, if firms can infer preferences from purchase histories, the environment converges to one with perfect information (i.e., $\hat{v}_{ij} = v_{ij}$ for all i, j).

A key modeling choice is to allocate ads through dynamic auctions. This both reflects the microstructure of digital advertising and ensures tractability under multi-homing: In static settings, symmetric monotone bidding equilibria break down when a firm can enter multiple auctions for the same consumer. Spreading competition over time guarantees that each firm participates in at most one auction per consumer at a time, supporting equilibrium existence. Similar difficulties arise in static Walrasian formulations when platforms have asymmetric data. Supplemental Appendix G discusses these issues in more detail.

Beyond these technical benefits, the dynamic structure separates short-run from long-run effects, captures investment incentives more faithfully, and delivers steady-state predictions that are independent of initial conditions.

IV. Equilibrium of the Baseline Model

The equilibrium derivation proceeds in five steps. Proofs are in the Appendix. First, solving the attention allocation problem (14) yields

$$(15) \quad x_{kt} = \frac{[\nu(a_{kt}) q_{kt}]^{\epsilon-1}}{\int_{\mathbb{K}} [\nu(a_{zt}) q_{zt}]^{\epsilon-1} dz}, \quad k \in \mathbb{K}.$$

Thus, attention depends on effective quality—reflecting both service quality and ad load—relative to competitors.

Second, given (15), each platform chooses its ad rate to maximize flow profit:

$$(16) \quad A = \arg \max_{a_{kt}} \pi_{\mathbb{K}t} a_{kt} \frac{[\nu(a_{kt}) q_{kt}]^{\epsilon-1}}{\int_{\mathbb{K}} [\nu(a_{zt}) q_{zt}]^{\epsilon-1} dz} = \arg \max_a a \nu(a)^{\epsilon-1}.$$

This choice is *static* because no individual platform has measurable effects on consideration sets. CES implies *constant ad rates* determined only by primitives $\nu(\cdot)$ and ϵ .

Third, setting $A_t = A$, I solve (4) and (6) *analytically* for M_t and H_t , under symmetric and monotone bidding strategies.¹⁷

Fourth, given *arbitrary paths* for M_t and H_t , I solve (8)–(10) *explicitly* for bid functions $B_t(\cdot)$, verifying monotonicity and uniqueness.

Fifth, from $B_t(\cdot)$, I compute the average ad price $\pi_{\mathbb{K}t}$ and solve (12), yielding a simple boundary value problem that characterizes investment and quality paths.

Together, these steps deliver a tractable characterization of the full equilibrium transition path. The full system is reported in the Appendix, with most variables expressed analytically. Theorem 1 summarizes steady-state properties. In equilibrium, attention-market outcomes and product-market outcomes are jointly determined through consideration sets that link the two markets. In particular, ad revenue (part 4), investment and quality (part 5), and aggregate consumption (part 7) reflect this interaction. Capturing this joint determination allows policy and welfare to be evaluated within the model.

THEOREM 1: *Suppose A uniquely solves $\max_a a \nu(a)^{\epsilon-1}$. If $A/\lambda_f < J$ and $\epsilon - 1 < 1/\varphi$, then a unique equilibrium exists for any feasible initial condition (M_0, H_0, q_0) , characterized by equations (29)–(33) in the Appendix.¹⁸ The equilibrium converges to a steady state in which:*

1. *Each platform displays ads at rate A .*
2. *The mass M of products in Ω_{it} is A/λ_f .*

¹⁷To compute M_t and H_t , it is not necessary to first solve for attention allocation or platform quality. This simplicity relies on platform symmetry. With asymmetric data (Section VIII), the evolution of H_t depends on how attention is allocated, introducing an additional fixed point into the equilibrium characterization.

¹⁸An initial condition is *feasible* if $M_0 dH_0 \leq J dG$, since there cannot be more firms with a given expectation in Ω_{it} than exist in $\mathbb{J}$.

3. The c.d.f. H satisfies $H(\cdot) = H^c(\cdot)^N$, where H^c solves

$$MH^c(\hat{v})^N + (J - M)H^c(\hat{v}) = JG(\hat{v}), \hat{v} \in [0, \infty).$$

4. Total ad revenue is $\pi_{\mathbb{K}}A$, where $\pi_{\mathbb{K}}$ is the average ad price:

$$(17) \quad \pi_{\mathbb{K}} = \pi_{\mathbb{J}} \int_0^\infty \frac{1 - NH^c(\hat{v})^{N-1} + (N-1)H^c(\hat{v})^N}{\rho + \lambda_f + \lambda_e(\hat{v})} d\hat{v}.$$

Here, $\lambda_e(\hat{v})$ is the entry intensity into consideration sets (see (11)), and $\pi_{\mathbb{J}}$ scales firms' flow profits (see (2), with $\mu_{\Omega_i} = \mu_H$).

5. Each platform invests at rate

$$(18) \quad \ell_{\mathbb{K}} = \frac{\varphi \delta \pi_{\mathbb{K}} A (\epsilon - 1)}{K(\rho + \delta)},$$

and has quality $q = (1/\delta) \ell_{\mathbb{K}}^\varphi$.

6. Consumers allocate attention as in (15).

7. Aggregate platform consumption is $X_{it} = K^{1/(\epsilon-1)} \nu(A)q$, and aggregate product consumption is $C_{it} = \frac{\sigma-1}{\sigma} I(M\mu_H)^{1/(\sigma-1)}$.

The condition $\epsilon - 1 < 1/\varphi$ ensures concavity of platforms' investment problem; under the reverse inequality, no equilibrium exists. Similarly, $A/\lambda_f < J$ ensures consideration sets remain incomplete so that (6) applies.

Three implications of Theorem 1 are especially relevant for the analysis that follows, highlighting the structural sources of platform market power and the role of data.

Remarks:

1. **Elasticity of Attention:** Parts 5 and 6 show that the key parameter governing platform market power in attracting attention is ϵ . As $\epsilon \rightarrow 1$, platforms capture equal shares of attention regardless of quality or ad load. Holding all else constant, as ϵ falls, investment declines (part 5) while the ad display rate $A = \max_a a\nu(a)^{\epsilon-1}$ rises (part 1).
2. **Search Frictions:** Part 4 shows that search frictions generate market power in the ad market: As the entry intensity $\lambda_e(\cdot)$ rises pointwise, ad revenue falls and eventually vanishes. As $A/\lambda_f \rightarrow J$, the economy converges to a classical frictionless benchmark in which consideration sets contain all firms, platforms earn no rents, and therefore do not invest in quality.
3. **Role of Data:** Part 3 shows that data shapes the composition H of consideration sets through G . Through its effect on H , data reallocates demand toward higher-valuation firms, thereby altering ad revenue (part 4), investment incentives (part 5), aggregate consumption (part 7), and ultimately welfare.

V. Policy Experiments

Data policies and interoperability mandates are central to current debates on digital markets. This section shows that their effects can be *counterintuitive and time dependent*: Static predictions can reverse in transition, and interventions often trade off product-market efficiency against platform investment and quality. The analysis is intentionally stylized to highlight mechanisms and provide intuition for market conditions when these patterns arise. Throughout, formal results apply on the parameter domain where a unique equilibrium exists.

A. Data

I first consider data policies, analyzing steady state before turning to transition dynamics. Such policies change the informativeness of the data platforms supply to firms for bidding in ad auctions. An example is a privacy law that restricts the granularity of the data platforms can collect on users.

Data informativeness ranges from uninformative—when all firms' posterior expectations equal the prior mean μ_F —to perfectly informative, when posterior expectations coincide with consumers' true values ($G = F$). Intermediate cases are ordered by the mean-preserving spread order (Blackwell 1953).¹⁹

Steady State.—Proposition 2 shows that banning data use (or ad targeting) *raises* ad revenue and platform quality but lowers product consumption in steady state.²⁰

PROPOSITION 2: *In steady state, as data becomes uninformative (i.e., along any sequence of continuous distributions G converging pointwise to the degenerate distribution at μ_F), ad revenue and platform quality converge to their suprema, while aggregate product consumption converges to its infimum, over all continuous distributions G with mean μ_F . Moreover, aggregate product consumption is monotone in informativeness.*

Surprisingly, ad revenue is *highest* when data is *uninformative*. A data ban therefore *raises* platform *profits*.²¹ Three forces drive the result:

1. **Fixed Total Firm Profits:** Under CES demand, prices are fixed and budgets bind, so total firm profits, gross of ad costs, equal I/σ regardless of G .
2. **Full Extraction:** With uninformative data, Bertrand competition in auctions extracts these profits fully.
3. **Information Rents:** Informative data creates bidder information rents that reduce ad revenue.

¹⁹A c.d.f. G is a mean-preserving spread of a c.d.f. $\hat{G}$ if $\int_{-\infty}^x G(t) dt \leq \int_{-\infty}^x \hat{G}(t) dt, \forall x$ with equality when $x = \infty$.

²⁰Because G is assumed continuous to rule out ties, the case of uninformative data is formalized in a limiting sense in Proposition 2.

²¹Platform profits net of investment are monotone in ad revenue.

The fixed-profit force is a stark implication of CES demand. The broader mechanism, however, is product-market competition: When data improves, each firm internalizes that it faces rivals who are likewise better-matched for any given consumer. In the CES setting this force makes total firm profits exactly invariant to G ; more generally, competition reduces the surplus firms retain from better targeting.²² Provided this effect is sufficiently strong, the finding that less informative data may raise ad revenue does not hinge on profits literally being fixed.

Beyond the case of uninformative data, analytical monotone comparative statics are difficult to establish. Nevertheless, numerical results indicate that more informative data typically reduces ad revenue.

The intuition for this finding is as follows. Although more informative data raises the average value of winning firms—an effect that tends to increase ad revenue since bids are monotone—three forces push in the opposite direction as seen from the expression for ad revenue in part 4 of Theorem 1.

1. **Higher Match Rates for High-Value Firms:** More informative data raises the entry rates $\lambda_e(\hat{v})$ of firms with relatively higher $\hat{v}$.
2. **Lower Average Value Outside Consideration Sets:** The average value μ_H^c of firms bidding in auctions falls.
3. **Lower Profit Coefficient:** As μ_H rises, the coefficient $\pi_{\mathbb{J}}$ on firms' profits falls.

In steady state these forces often dominate, causing ad revenue to decline. This is so, particularly because of the third effect. To see why, note that a firm's expected flow profit from selling to a consumer is

$$\pi_{\mathbb{J}}\hat{v} = \frac{I\hat{v}}{\sigma M\mu_H}.$$

More informative data raises the $\hat{v}$ of winning firms, but it also raises μ_H , the average $\hat{v}$ in consideration sets. In steady state these coincide, so improvements in match quality do not increase winners' expected profits. This is another expression of the fixed-profits force.

Because auctions are second price, ad revenue depends on the expected profit of the second-highest bidder rather than that of the winner. Specifically, it depends on the ratio of the *second*-highest $\hat{v}$ in auctions to μ_H . This ratio is maximized at one when data is uninformative and typically—though not universally—declines as data becomes more informative.

Transition Dynamics.—The steady-state analysis suggests that transition dynamics matter because some key forces unfold only gradually as the composition of

²² Advertising's share of GDP has remained remarkably constant despite major changes in advertising technologies since 1925. As Chemi (2014) notes, "the pie is not growing...the easiest way to make more money is to steal larger slices of the pie." Silk and Berndt (2021) find that advertising's share of GDP declined slightly during the period 2000–2018 when digital advertising grew in dominance. I revisit this phenomenon in Section VIII.

firms in consideration sets adjusts. Following an increase in data informativeness, two mechanisms—the decline in the average expected value of bidders outside consideration sets, μ_{H^c} , and the fall in the profit coefficient $\pi_{\mathbb{J}}$ —occur over time. As a result, ad revenue may rise on impact even though it ultimately falls in the long run.

Figure 2 illustrates this pattern following an unanticipated increase in data informativeness at $t = 1$, where G shifts from $U[0.2, 0.8]$ to $U[0, 1]$ as the economy transitions between steady states.²³ This change may represent, for example, a relaxation of privacy laws that allows platforms to target ads with more granular data.

Ad revenue *initially rises* but eventually falls below its starting level. Investment and quality decline steadily, while product consumption rises monotonically. Investment falls despite the temporary increase in ad revenue because the increase is short-lived and platforms are forward-looking.

In this example, data liberalization can appear beneficial for platforms *on impact* but ultimately reduces platform profits *in the long run*. Because platform quality and product consumption move in opposite directions, the welfare effect is ambiguous and depends on $u(\cdot)$.

I close with three remarks.

Remarks:

1. **When These Patterns Arise:** The patterns in Figure 2 tend to arise in numerical examples when firms are less patient and platforms are more patient. Less patient firms put less weight on the future decline in profits caused by better-matched competitors, so bids respond more strongly to the immediate improvement in targeting. More patient platforms put more weight on the eventual decline in ad revenue, so investment falls despite the short-run increase in ad revenue.
2. **Private versus Collective Incentives:** Though platforms may collectively be worse off when data improves for *all* platforms, *individual* platforms typically benefit from private access to more informative data. I provide an illustrative example in Section VIII.
3. **Settings with Personalized Pricing:** In alternative models where prices are personalized in equilibrium, total firm profits may vary with data informativeness. Whether the qualitative comparative statics here are reinforced or mitigated in such settings is an open question.²⁴

²³Figure 6 in the Appendix shows the specific garbling of new data into old data corresponding to this change in G . The details of the garbling—i.e., the joint distribution of old and new expectations—do not affect the steady state, but they matter for transition dynamics because they determine the initial distribution of expectations in consideration sets after the shock.

²⁴Rhodes and Zhou (2024) show that under “full market coverage,” personalized pricing can reduce firm profits, reinforcing the direction of the steady-state results here.

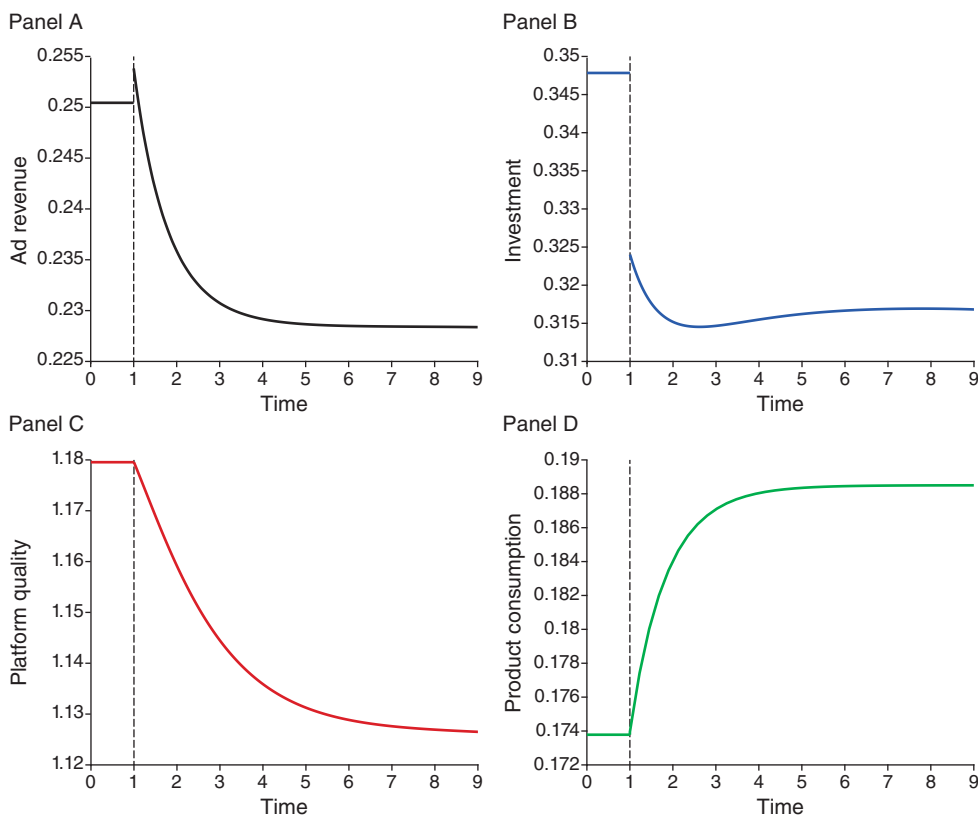


FIGURE 2. TRANSITION DYNAMICS OF INCREASED DATA INFORMATIVENESS

Notes: Transition between steady states following a data shock at $t = 1$, where the signal distribution changes from $G = U[0.2, 0.8]$ to $G = U[0, 1]$ (corresponding to the garbling in Figure 6). Parameters: $\rho = 0.1$, $\epsilon = 4/3$, $\lambda_f = 1$, $K = 0.1$, $J = 1$, $\sigma = 3$, $N = 5$, $I = 1$, $\varphi = 0.5$, $\delta = 0.5$, and $\nu(a) = 1 - 7.5a$.

B. Interoperability

Interoperability mandates aim to reduce user lock-in and promote platform competition by making attention more *elastic* to platform quality. For example, consider a content-interoperability rule requiring platforms to allow content sharing across services. Because content becomes more similar across platforms, differences in interface quality or ad load play a more prominent role in attracting attention.

In practice, interoperability mandates may involve technological changes to platforms that alter the aggregator X or even the utility function u on a case-by-case basis. The analysis below remains agnostic about these details and assumes only that *attention* retains its CES shape and becomes *more elastic*—captured by a higher ϵ —after the policy.²⁵ Interpreted broadly, the analysis is therefore relevant for any pro-competitive intervention that raises the elasticity of attention.

²⁵For instance, the aggregator X may be scaled or translated following the mandate.

Steady State.—The following proposition establishes steady-state comparative statics of an increase in ϵ .

PROPOSITION 3: *An increase in ϵ reduces steady-state product consumption. Moreover, holding the steady-state CDFs H and H^c fixed while allowing M to adjust according to $M = A/\lambda_f$, the partial-equilibrium effect of an increase in ϵ is to raise steady-state ad revenue and platform quality.*

In general, H and H^c also vary with ϵ through its effect on M (part 3 of Theorem 1). In Supplemental Appendix C, I give a sufficient condition for the full-equilibrium effects on ad revenue and platform quality to remain positive: G admits a density that is positive on the interior of its support, with $\inf \text{supp } G = 0$, and a decrease in M decreases H^c in the convex-transform order²⁶ (Condition 1, Proposition 6). This condition holds, for example, when G is a scaled power distribution, $G(x) = (x/b)^\alpha$ on $[0, b]$ with $\alpha \in (0, 1]$ (Lemma 12). The uniform distribution on $[0, b]$ is the special case $\alpha = 1$.

The mechanism behind Proposition 3 is as follows.

1. **Lower Ad Load:** More elastic attention induces platforms to reduce the ad display rate A and invest more in quality q .
2. **Smaller Consideration Sets:** Lower A shrinks consideration sets ($M = A/\lambda_f$), reducing product consumption C .
3. **Higher Ad Prices:** Holding H and H^c fixed, the ad price $\pi_{\mathbb{K}}$ rises sufficiently that total ad revenue $\pi_{\mathbb{K}}A$ increases despite the decline in A . As seen from (17) in Part 4 of Theorem 1, $\pi_{\mathbb{K}}$ rises because (i) the firm profit coefficient $\pi_{\mathbb{J}}$ scales inversely with A via $M = A/\lambda_f$, and (ii) lower A reduces match rates $\lambda_e(\cdot)$.

Step 3 holds H and H^c fixed. In Supplemental Appendix C, I show that both H and H^c decrease pointwise. Thus the distribution H^c of expected values of bidders improves in the sense of first-order stochastic dominance, raising ad revenue. However, the distribution H of expected values in consideration sets also improves, raising μ_H and thus reducing $\pi_{\mathbb{J}}$ —a force that lowers ad revenue. Condition 1 in Supplemental Appendix C ensures that the net effect on ad revenue is positive.

Forces that raise the ad price operate gradually, through the dynamics of M_t and H_t^c , whereas the effect on the ad rate A is immediate. As a result, ad revenue may *decline* initially even when its new steady-state level is higher, making transition dynamics important.

Transition Dynamics.—A one-time 1 percent increase in ϵ at $t = 1$ produces the transition path shown in Figure 3.

²⁶ Writing F_0 and F_1 for H^c before and after the decrease in M , this means that $F_0^{-1} \circ F_1$ is convex: The decrease in M makes the distribution of values outside consideration sets less right-skewed. Supplemental Appendix C gives the formal definition.

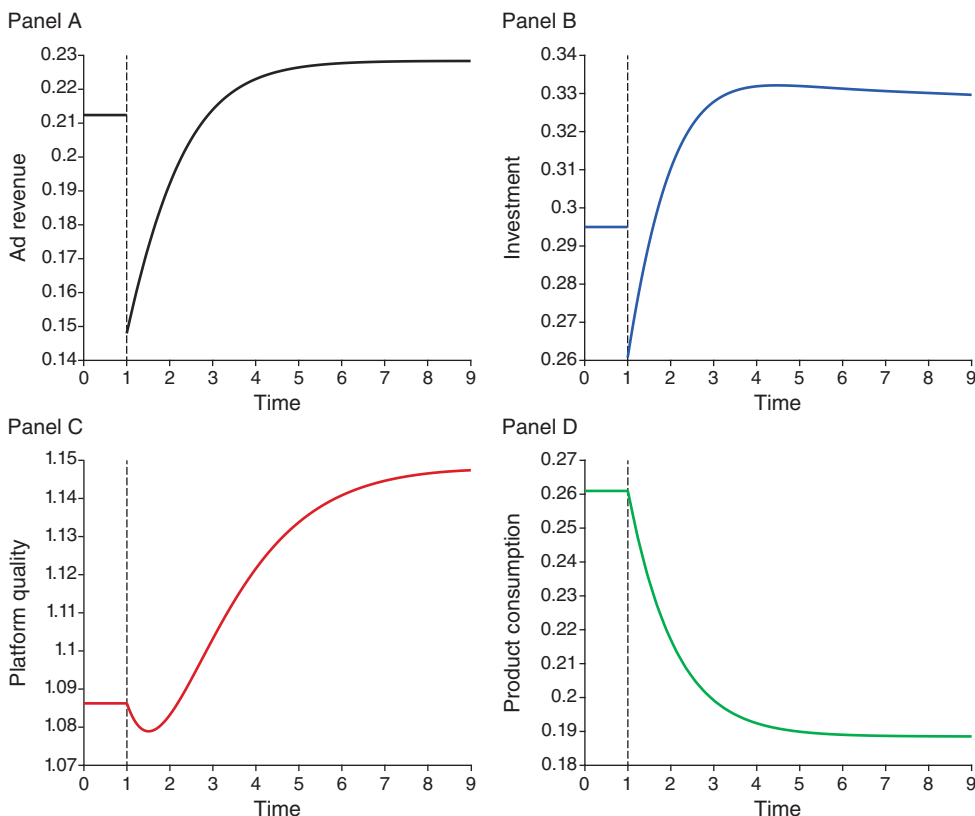


FIGURE 3. TRANSITION DYNAMICS OF INCREASED ATTENTION ELASTICITY

Notes: Transition between steady states following an unanticipated shock to ϵ from $4/3$ to $(4/3)(1.01)$ at $t = 1$. Parameters: $\lambda_f = 1$, $K = 0.1$, $J = 1$, $\rho = 0.1$, $\sigma = 3$, $N = 5$, $G = U[0, 1]$, $I = 1$, $\varphi = 0.5$, $\delta = 0.5$, $\nu(a) = (1 - 0.8395a^{.01})^{63.1472}$.

Interoperability depresses ad revenue and investment *at first*, yet raises them *in steady state*. In the long run, product consumption declines while platform quality improves. Thus, as with an increase in data informativeness, short-run and long-run effects differ and the welfare effect is ambiguous and depends on $u(\cdot)$.

Remarks:

1. **When These Patterns Arise:** The patterns in Figure 3 tend to arise in numerical examples when nuisance costs are steeper and firms and platforms are less patient. Steeper nuisance costs make the drop in ad load larger on impact. Less patient firms place less weight on the future decline in consideration sets, so bids do not rise enough initially to offset the lower ad load. As a result, ad revenue falls more sharply in the short run. Less patient platforms, in turn, put less weight on the eventual rebound in ad revenue when choosing investment.

2. **Alternative Entry Channels:** If there is an alternative entry channel into consideration sets, higher ϵ might *reduce* steady-state ad revenue because M may no longer shrink proportionally with A .²⁷

VI. Welfare Analysis in General Equilibrium

So far, I have analyzed policy interventions without characterizing the underlying sources of inefficiency. This section closes the economy so that welfare corresponds to consumer surplus, and then compares the decentralized equilibrium with the planner's allocation. Specializing to Cobb–Douglas utility u , I show that equilibrium ad display and investment rates may lie on either side of the efficient levels, and I derive a sufficient statistic that signs the investment distortion. Although this statistic relies on monopolistic competition among platforms, the planner's solution extends directly to oligopoly settings.

A. General Equilibrium

To close the baseline model, I endogenize consumers' incomes. Each consumer supplies L units of labor inelastically at each date. Labor is the only input: Platforms use it for investment, and firms use it for production. Thus ℓ_{kt} now denotes the labor hired by platform k for investment. One unit of output requires one unit of labor, so firms' marginal cost equals the wage, which I normalize to one.

Labor market clearing at time t requires

$$L = \int_{\mathbb{K}} \ell_{kt} dk + \int_{\mathbb{J}} \ell_{jt} dj,$$

where

$$\ell_{jt} = \int_{\mathbb{I}} c_{ijt} \mathbf{1}_{\{j \in \Omega_{it}\}} di$$

is firm j 's total labor demand.

Consumers collectively own all firms and platforms in equal shares. Since platforms earn only ad revenue and all other costs are wages, income I_t equals aggregate firm revenue:

$$I_t = \int_{\mathbb{J}} p_{jt} \ell_{jt} dj.$$

This identity also follows from product-market clearing.

An equilibrium is defined as in the baseline model, augmented with the income process $\{I_t\}$ and the requirement that labor and product markets clear at all dates.

Proposition 4 provides the general-equilibrium counterpart to Theorem 1.

²⁷This logic does not apply to word-of-mouth referrals if initial discoveries must originate from advertising. In that case, word-of-mouth is a downstream consequence of advertising rather than a truly separate entry channel.

PROPOSITION 4: *Theorem 1 applies as stated, except with income $I_t = \frac{\sigma}{\sigma-1}(L - K\ell_{\mathbb{K}t})$ in place of I . The equilibrium converges to a steady state in which each platform invests at rate*

$$(19) \quad \ell_{\mathbb{K}} = \frac{\varphi \delta^{\frac{\sigma}{\sigma-1}} \hat{\pi}_{\mathbb{K}} A (\epsilon - 1)}{\rho + \delta + \varphi \delta^{\frac{\sigma}{\sigma-1}} \hat{\pi}_{\mathbb{K}} A (\epsilon - 1)} \frac{L}{K},$$

where $\hat{\pi}_{\mathbb{K}} \equiv \pi_{\mathbb{K}}/I$ denotes the average ad price per unit of income.

The comparative statics from Propositions 2 and 3 also extend to this general equilibrium setting, with $\hat{\pi}_{\mathbb{K}}A$ replacing $\pi_{\mathbb{K}}A$ where relevant (Supplemental Appendix D.2).

I next characterize the planner's allocation before comparing it with the decentralized equilibrium.

B. Planner's Problem

The planner chooses ad display and investment rates to maximize welfare, taking as given that consumers choose demands to maximize flow utility and firms set prices and bids to maximize discounted profits. The planner is constrained to treat platforms symmetrically, observes the same consumer data as platforms, and cannot impose technological changes such as interoperability. Even so, this benchmark is valuable for evaluating data and interoperability policies, whose welfare effects depend on the wedge between the decentralized and planner allocations (see the remark after Corollary 2.2). Although the planner could also set firms' prices or bids, these choices are already efficient.²⁸

Given (M_0, H_0, q_0) , the planner solves

$$(20) \quad \max_{\{\ell_{\mathbb{K}t}\}, \{A_t\}} \int_0^{\infty} e^{-\rho t} u(C_t, X_t) dt$$

subject to

$$C_t = (L - K\ell_{\mathbb{K}t}) (M_t \mu_{H_t})^{\frac{1}{\sigma-1}},$$

$$X_t = K^{\frac{1}{\epsilon-1}} \nu(A_t) q_{\mathbb{K}t},$$

with M_t and H_t following (4)–(6), and $q_{\mathbb{K}t}$ following (13) given $\ell_{\mathbb{K}t}$.²⁹

I focus on steady-state solutions.³⁰

²⁸Firms charge markups, but all produce identical quantities, which is efficient by concavity of the CES aggregator and ex ante symmetry. Bids are efficient because they are monotone.

²⁹In C_t , the term $L - K\ell_{\mathbb{K}t}$ is labor available for production.

³⁰I conjecture that, under standard regularity conditions, any solution converges to the steady state, but do not pursue this here.

DEFINITION 2: A steady-state solution of the planner's problem consists of initial conditions (M^*, H^*, q^*) and constants A^* and $\ell_{\mathbb{K}}^*$ such that the planner sets $A_t = A^*$ and $\ell_{\mathbb{K}t} = \ell_{\mathbb{K}}^*$, with $M_t = M^*$ and $H_t = H^*$ for all t .

For clarity, I assume

$$(21) \quad u(C_t, X_t) = C_t^{1-\tau} X_t^\tau,$$

with $\tau \in (0, 1)$ for the remainder of this section.

Theorem 2, proven in Supplemental Appendix D, characterizes steady-state investment for arbitrary ρ and steady-state ad display in the limit as $\rho \rightarrow 0$.³¹

THEOREM 2: Let u be given by (21). Any steady-state solution has investment

$$(22) \quad \ell_{\mathbb{K}}^* = \frac{\varphi \delta \frac{\tau}{1-\tau}}{\rho + \delta + \varphi \delta \frac{\tau}{1-\tau}} \frac{L}{K}.$$

Suppose a steady-state solution exists for all ρ near zero. Let $A^*(\rho)$ denote the corresponding ad display rate. Then

$$(23) \quad \lim_{\rho \rightarrow 0} A^*(\rho) = \arg \max_a \left[a \mu_H(a) \right]^{\frac{1-\tau}{\sigma-1}} \nu(a)^\tau,$$

whenever the right-hand side is well defined. Here $\mu_H(a)$ is the steady-state average quality of firms in consideration sets when all platforms use ad display rate a .

COROLLARY 2.1: If the planner controls investment but not ad display, the unique steady-state choice of investment remains (22).

C. Planner versus Laissez-Faire

I now compare the planner's allocation with the decentralized equilibrium.

Investment.—Comparing (19) with (22) yields a simple sufficient statistic.

PROPOSITION 5: For a given τ , the deviation $\ell_{\mathbb{K}} - \ell_{\mathbb{K}}^*$ is increasing in

$$(24) \quad \frac{\sigma}{\sigma-1} \hat{\pi}_{\mathbb{K}} A (\epsilon - 1) - \frac{\tau}{1-\tau}.$$

Equilibrium investment is too high (too low) when (24) is positive (negative). Either case can arise, since $\hat{\pi}_{\mathbb{K}}$ does not depend on τ (see (43), Supplemental Appendix D).

This statistic depends on four interpretable objects: (i) the product-market markup $\sigma/(\sigma-1)$; (ii) ad revenue per unit of income $\hat{\pi}_{\mathbb{K}} A$; (iii) the elasticity

³¹ The ad display rate for general ρ can also be derived using Pontryagin's maximum principle but is less transparent to state.

$\epsilon - 1$ of attention with respect to platform quality; and (iv) the utility weight τ on platform consumption.³²

These reflect distinct distortions. Markups capture market power, which shifts labor toward platform investment. Ad revenue reflects business-stealing motives whose strength is governed by the elasticity $\epsilon - 1$. The utility weight appears because platforms, which charge zero prices, do not internalize the consumer-surplus gains from higher quality whereas the planner does.

A proportional tax or subsidy on ad revenue can correct these distortions.

COROLLARY 2.2: *A proportional tax on platform ad revenue at rate*

$$1 - \frac{\tau}{1 - \tau} \frac{\sigma - 1}{\sigma} \frac{1}{\hat{\pi}_{\mathbb{K}} A (\epsilon - 1)},$$

rebated lump-sum to consumers, restores $\ell_{\mathbb{K}}$ to the efficient level (22). A negative rate corresponds to a subsidy financed by consumers.

Remark: Condition (24) also helps evaluate policy changes. The comparative statics for $\pi_{\mathbb{K}} A$ in Propositions 2 and 3 extend to $\hat{\pi}_{\mathbb{K}} A$ in general equilibrium. Thus a ban on data use reduces investment efficiency and welfare whenever investment is initially too high. Analogous logic applies to changes in ϵ , such as those induced by interoperability.

Ad Display Rate.—From (16) and (23) the equilibrium ad rate maximizes

$$a\nu(a)^{\epsilon-1},$$

whereas, as $\rho \rightarrow 0$, the planner's rate maximizes

$$\left[a\mu_H(a) \right]^{\frac{1-\tau}{\sigma-1}} \nu(a)^{\tau}.$$

Hence the equilibrium rate may be above or below the planner's. As an illustration, suppose ν is continuous and satisfies $\nu(\bar{a}) = 0$ for some $\bar{a} > 0$.³³ Then, as $\epsilon \rightarrow 1$, the equilibrium rate converges to $\bar{a}$ whereas the planner's rate does not, so the equilibrium rate is inefficiently high.

Platforms internalize the nuisance cost of ads only through its effect on attention, since zero prices prevent platforms from capturing the consumer-surplus gains from lower ad loads directly. When ϵ is small and attention is inelastic, this internalization is weak, leading to excessive ads.

Conversely, the equilibrium rate may be too low. As $\tau \rightarrow 0$, the planner's optimal rate converges to $\bar{a}$, while the equilibrium rate remains unchanged. In this case

³²Since τ appears in the sufficient statistic (24), its empirical counterpart matters for evaluating investment distortions. In principle, one could estimate τ and ϵ jointly by an experiment that charges users for access to platform services.

³³The baseline assumption $\nu > 0$ was made for simplicity, since demand (15) is undefined if almost all platforms choose a with $\nu(a) = 0$. Theorem 1 extends to the case in which ν vanishes above $\bar{a} > 0$, provided we restrict attention to equilibria in which each platform sets ad display below $\bar{a}$.

the planner places little weight on platform consumption and primarily values product consumption, increasing the ad display rate to expose consumers to more products. Platforms do not internalize this channel and may choose a lower rate than the planner.

VII. Duopoly

This section shows that the framework can also accommodate strategic behavior in a duopoly setting and develops a simple numerical algorithm for computing Markov equilibrium. For tractability, ad display rates are taken as exogenous. I use the duopoly model to assess the robustness of Section V's findings, particularly since interoperability mandates often target large platforms. I remark on the case of endogenous ad rates at the end of the section.

A. Setup

There are two platforms, $k \in \mathbb{K} \equiv \{1, 2\}$. Platform k 's quality evolves according to

$$(25) \quad dq_{kt} = (\ell_{kt}^\varphi - \delta q_{kt})dt + \eta q_{kt} dB_{kt}, \quad q_{k0} = q_0,$$

where B_{1t} and B_{2t} are independent Brownian motions. The law of motion mirrors the baseline model, but with the addition of noise. A small amount of noise is useful for numerical stability and, as in related dynamic duopoly models, appears to help ensure existence of a Markov equilibrium.³⁴ In the numerical examples, I set $\eta = 0.12$.

To isolate strategic effects along the investment margin, both platforms display ads at a common exogenous rate A . For the particular choice of A given by (16), this assumption admits an interpretation in which platforms set ad display *as if* they were individually small on this margin, as in the baseline. All other elements of the baseline environment are retained.

B. Markov Equilibrium

I focus on Markov equilibrium where strategies depend only on the two quality levels and time. Let $V_k(q_1, q_2, t)$ denote platform k 's value function. Suppressing arguments, value functions satisfy the coupled HJB system

$$(26) \quad \rho V_k = \sup_{\ell_k} \mathcal{L}^{(\ell_1, \ell_2)} V_k + \pi_{\mathbb{K} \setminus k} A \frac{q_k^{\epsilon-1}}{q_1^{\epsilon-1} + q_2^{\epsilon-1}} - \ell_k, \quad k \in \mathbb{K},$$

³⁴ See Budd, Harris, and Vickers (1993) and Harris, Howison, and Sircar (2010). In such settings, deterministic formulations often fail to admit Markov equilibrium because, under certain Markov strategy profiles, an arbitrarily small lead can generate large payoff gains for a player. Introducing noise can restore equilibrium existence by softening these leader—follower dynamics. Consistent with this, Budd, Harris, and Vickers (1993), who analyze a similar model, claim in their Section III that any nontrivial amount of noise ensures equilibrium existence. Given the close connection between our models (see Supplemental Appendix E), I conjecture the same mechanism applies here.

where $\mathcal{L}^{(\ell_1, \ell_2)}$ is the infinitesimal generator of the joint quality process: For any sufficiently smooth $F : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$\begin{aligned} \mathcal{L}^{(\ell_1, \ell_2)} F = & F_t + (\ell_1^\varphi - \delta q_1) F_{q_1} + (\ell_2^\varphi - \delta q_2) F_{q_2} + \frac{1}{2} \eta^2 q_1^2 F_{q_1 q_1} \\ & + \frac{1}{2} \eta^2 q_2^2 F_{q_2 q_2}. \end{aligned}$$

Since all equilibrium objects other than investment coincide with those in the baseline model, solving for a Markov equilibrium reduces to finding value functions and investment policies that satisfy this system.

Algorithm Overview.—I solve the HJB system using an implicit upwind finite-difference scheme. In steady state, value functions and policies depend on (q_1, q_2) but not on t .

1. **Steady State:** Fix the steady-state ad price from Theorem 1. From an initial guess of value functions, iterate (26) backward in time, updating value functions and controls until convergence to steady state.
2. **Transition Path:** Using the steady-state value functions as the terminal condition, iterate (26) backward from a large terminal time to compute value functions and controls along the transition path.

Details and theoretical foundations are discussed in Supplemental Appendix E.

C. Policy Experiments under Duopoly

I revisit the policy experiments from Section V to assess the robustness of the results to strategic interaction in investment. I initialize both platforms' quality levels at the point where their drifts are zero in steady-state and plot the transition following a parameter shock along the path where *realized* Brownian shocks are zero. Because the equilibrium is symmetric, the two platforms share a common quality trajectory along this path. Although transition dynamics may vary across realized Brownian paths, I plot this trajectory because it is representative given the modest volatility parameter $\eta = 0.12$.

Figure 4 presents two experiments. Panels A and B examine a data shock from $G = U[0.2, 0.8]$ to $G = U[0, 1]$. Panels C and D examine an interoperability shock that raises ϵ (from $4/3$ to $(4/3)(1.01)$) and lowers the ad rate A (from 0.2 to 0.1). If platforms set ad display as in the baseline model (by ignoring strategic effects or the effects on consideration sets), this change in A would arise endogenously from the change in ϵ . The experiment thus isolates the *incremental effects* of strategic investment relative to the baseline model. I plot quality and investment paths. Consumption and ad revenue follow the same paths as in the baseline up to a constant scaling factor and are therefore omitted for brevity.

Across both experiments, the qualitative comparative statics mirror those in the monopolistically competitive case, showing that the key economic mechanisms are robust to strategic investment.

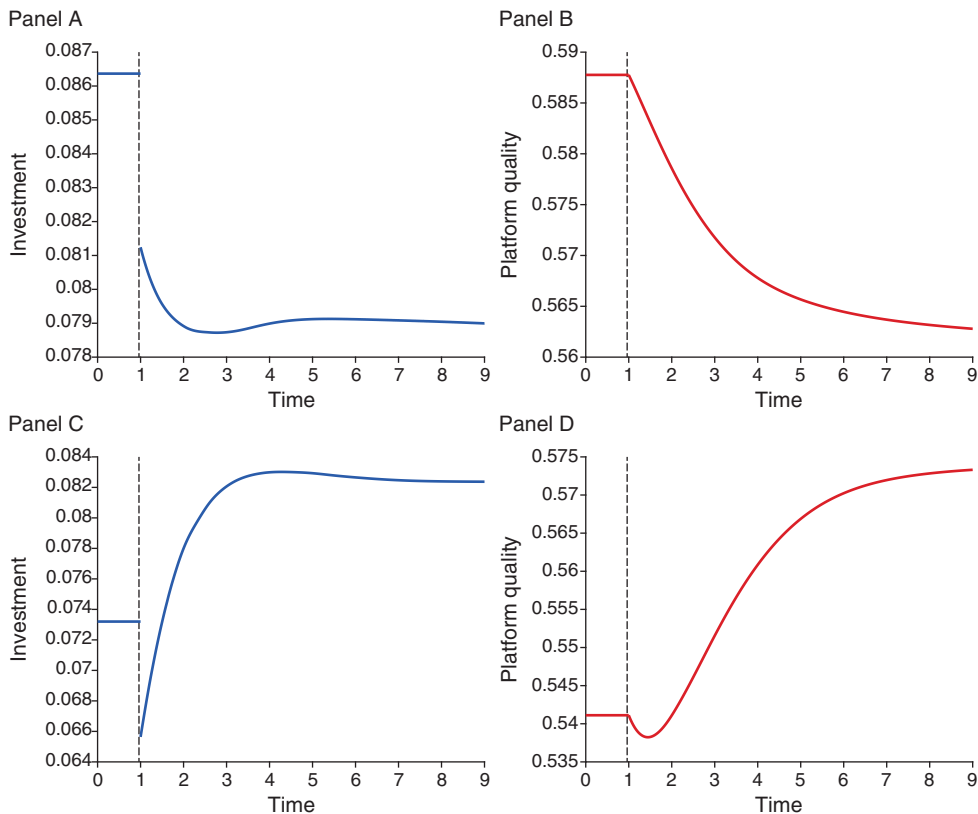


FIGURE 4. TRANSITION DYNAMICS UNDER DUOPOLY

Notes: Panels A and B show responses to a shock to G from $U[0.2, 0.8]$ to $U[0, 1]$, using parameters from Figure 2 except with $I = 10$. Panels C and D show responses to simultaneous shocks to ϵ from $4/3$ to $(4/3)(1.01)$ and to A from 0.2 to 0.1, using parameters from Figure 3 except with $I = 10$.

D. Comment on Endogenous Ad Display Rates

The analysis treats the ad display rate A as exogenous. Endogenizing ad display requires tracking the entire distribution H_t as a state variable, which complicates the equilibrium computation. A possible route is to instead model a continuum of duopolistic platform markets (for example via a nested CES structure). In that environment, ad display remains a static choice and no individual platform affects the deterministic law of motion of consideration sets. At the same time, platforms remain large under a narrower market definition, with strategic interaction in both investment and ad display. The algorithm developed here appears extendable to this setting particularly for steady state.³⁵

³⁵In such a setting, the interoperability counterfactual corresponds to raising the elasticity in each market, capturing the aggregate effect of many mandates (or a mandate applied across markets).

VIII. Data Asymmetry

So far, platforms have shared the same data, and we found that they may *collectively* prefer coarser information. I now extend the framework to allow data asymmetry between two groups of platforms, linking data informativeness to relative market shares and quality levels. In this environment, *individual* platforms often benefit from more granular data even when, collectively, platforms do not. Building on this, I offer a candidate mechanism for the puzzling decline in advertising's share of GDP during 2000–2018, precisely when *data-rich* digital platforms rose to prominence (Silk and Berndt 2021).

The dynamic auction structure is crucial for keeping the analysis here tractable, as discussed in Subsection IIIA. To my knowledge, relatively little work studies platform competition with asymmetric data.³⁶

A. Setup

There are two groups of platforms indexed by $z \in \{1, 2\}$, with masses m_z . As before, v_{ij} denotes consumer i 's value for firm j . When bidding on a platform in group z , firm j receives signal ζ_{zij} . I assume

$$(v_{ij}, \zeta_{1ij}, \zeta_{2ij}) \sim \mathcal{Q}$$

on $[0, \infty) \times \mathbb{R}^2$, independently across i and j . Let G be the joint c.d.f. of $(\zeta_{1ij}, \zeta_{2ij})$ with continuous density g . All other aspects of the baseline model are retained.

B. Equilibrium

With asymmetric data, a state variable is the *joint distribution* H_t of signals $(\zeta_{1ij}, \zeta_{2ij})$ in consideration sets. Its law of motion depends on the attention shares allocated across the two platform groups, which depend on platform investment, which depends on ad revenue, which depends on H_t —introducing an additional fixed point relative to the baseline model. Bidding strategies also differ by platform group. Despite this added structure, the model remains parsimonious, and I provide a simple algorithm to compute steady-state equilibrium.³⁷

Algorithm Overview.—A steady-state equilibrium is computed as follows:

1. Guess an attention share x_1 allocated to group 1 platforms.
2. Fixing x_1 , iterate the law of motion (48) for H_t forward until convergence.

³⁶See exceptions Ichihashi (2021); Greenwood, Ma, and Yorukoglu (2025); and Bergemann and Bonatti (2011). The latter two analyze perfect competition between traditional and digital media. In Greenwood, Ma, and Yorukoglu (2025), targeting distinguishes college from noncollege consumers; in Bergemann and Bonatti (2011), targeting ability is captured through the joint distribution of preferences and media use.

³⁷Details and referenced equations are in Supplemental Appendix F.

3. Compute steady-state bid functions and average ad prices via a contraction map (51).
4. Given ad prices, platform qualities and the implied attention share x_1 follow explicitly (53).
5. If the implied x_1 matches the guess, a steady-state equilibrium is found; otherwise update the guess and repeat.

By iterating over a grid of guesses for x_1 on $[0, 1]$, one can identify all steady-state equilibria and verify uniqueness.

C. A Puzzling Empirical Pattern

The model can reproduce the following stylized fact:

Perhaps the most puzzling feature...is that the rapid growth of digital advertising has occurred over a period during which the share of US economic activity (as measured by GDP) represented by total advertising expenditures has been in decline. (Silk and Berndt 2021, p. 4)

The mechanism mirrors Section V: More informative data raises the average expected value of firms in consideration sets, μ_H . Because firms' profits gross of advertising costs remain fixed, the surplus platforms can extract is limited. With greater information rents in auctions, *total* ad revenue can fall even as data-rich platforms gain market share. Though other explanations may exist, the model suggests that the data richness of digital platforms may itself have contributed to the decline in ad revenue.

To illustrate, suppose $v_{ij} = e^{Z_{ij}}$ with $Z_{ij} \sim N(0, \sigma_Z^2)$. There are two platform groups of equal mass. Firm j 's signals are

$$\zeta_{1ij} = Z_{ij} + \Delta u, \quad \zeta_{2ij} = Z_{ij} + u,$$

where $u \sim N(0, \sigma_u^2)$ and $0 \leq \Delta \leq 1$.³⁸ When $\Delta < 1$, group 1 platforms are data rich (digital), and group 2 platforms are data poor (traditional).

Figure 5 plots the steady-state share of ad revenue accruing to group 1 as a function of its data advantage $1 - \Delta$. The left-most point ($\Delta = 1$) corresponds to the case in which both groups are traditional media. Moving rightward, each point $\Delta < 1$ represents a steady state in which data-rich digital platforms have entered alongside traditional media. Both total ad revenue and its split across groups are invariant to the total mass of platforms, so entry alone *cannot* generate the patterns in the figure. Thus, the comparison between $\Delta = 1$ and any $\Delta < 1$ is consistent with *any* pattern of entry or exit, provided the new steady state contains equal masses of each group.

³⁸ Firm j never observes both signals simultaneously. When bidding on platform z , the firm observes only ζ_{zj} and does not know the realization of the other signal.

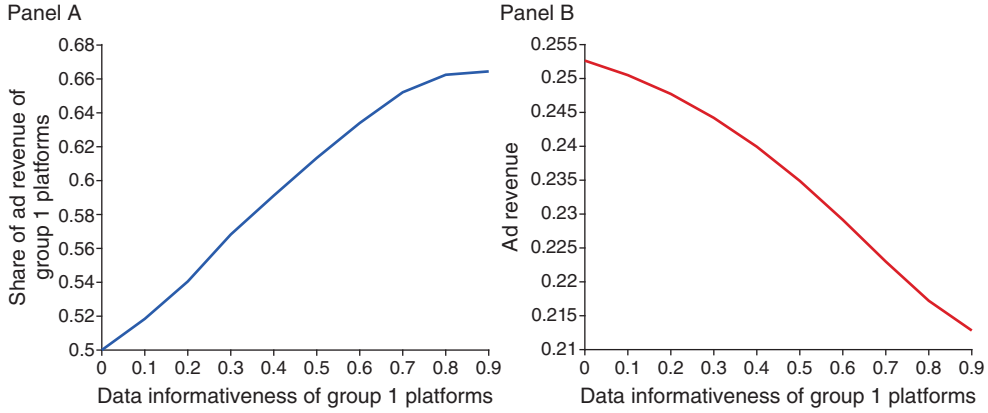


FIGURE 5. GROUP 1 SHARE OF TOTAL AD REVENUE AND TOTAL AD REVENUE

Notes: The group 1 share of ad revenue and total ad revenue are plotted as functions of the data advantage of group 1 platforms, measured by $1 - \Delta$, for parameter values $\sigma_Z^2 = 0.5$, $\sigma_u^2 = 2$, $I = 1$, $\rho = 0.1$, $A = 0.01$, $\lambda_f = 1$, $J = 0.1$, $N = 20$, $\epsilon = 1.6$, $\sigma = 3$, $\varphi = 0.75$, $\nu(a) = 1 - 62.5a$. Results hold for any platform mass K .

Consistent with the stylized fact, group 1 captures a rising share of ad revenue as its data advantage grows, yet *total* ad revenue declines. In this example, the higher share reflects both higher ad prices and higher attention shares. Thus, individual platforms gain from more informative data even when platforms as a whole do not.

IX. Further Extensions

I summarize four extensions of the baseline model to illustrate its ability to accommodate a wide range of market features. Although presented separately, these extensions can, in steady state, be combined within a unified framework. All four admit analysis in general equilibrium, where welfare has a natural interpretation, and extensions 1–3 also appear tractable in the duopoly and asymmetric-data environments.

1. **Network Effects:** In the first extension, found in Supplemental Appendix H, I introduce network effects by assuming that a platform's effective quality is $\eta(x_{kt})\nu(a_{kt})q_{kt}$, where $\eta(\cdot)$ increases in total attention $x_{kt} = \int_{\mathbb{I}} x_{ikt} di$. Analogous to (15), equilibrium attention satisfies

$$x_{kt} = \frac{[\eta(x_{kt})\nu(a_{kt})q_{kt}]^{\epsilon-1}}{\int_{\mathbb{K}} [\eta(x_{zt})\nu(a_{zt})q_{zt}]^{\epsilon-1} dz}, \quad k \in \mathbb{K}.$$

Since x_{kt} appears on both sides, to obtain explicit solutions I assume $\eta(x) = x^\zeta$ with $\zeta > 0$.

For any set $E_t \subseteq \mathbb{K}$ of positive mass, there is a solution of the form

$$x_{kt} = \frac{[\nu(a_{kt}) q_{kt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}}}{\int_{E_t} [\nu(a_{zt}) q_{zt}]^{\frac{\epsilon-1}{1-\zeta(\epsilon-1)}} dz} \mathbf{1}_{\{k \in E_t\}}, \quad k \in \mathbb{K}.$$

Thus network effects generate *equilibrium multiplicity*. Under the refinement that $E_t = \mathbb{K}$ at each t , a unique equilibrium exists. The equilibrium mirrors that of the baseline model but with a higher elasticity of attention to quality equal to $(\epsilon - 1)/[1 - \zeta(\epsilon - 1)] > \epsilon - 1$. This higher elasticity reduces equilibrium ad display rates and, under Condition 1, increases long-run investment. The refinement is natural when $\mathbb{K}$ is interpreted as a stable set of active platforms.

2. **Heterogeneous Platform Productivity:** The second extension, found in Supplemental Appendix I, allows platforms to differ in the productivity of their investments. I characterize full equilibrium dynamics. By varying the productivity distribution, the model can generate a broad class of nonatomic platform share distributions, enhancing its suitability for quantitative work and empirical applications.
3. **Zero Prices:** In the third extension, found in Supplemental Appendix J, I provide intuition for why zero prices for platform services may emerge endogenously in some environments. I allow platforms to charge only nonnegative prices. Negative prices may be infeasible if platforms cannot reliably distinguish bots from humans (Corrao, Flynn, and Sastry 2023).

The intuition is straightforward: If the marginal utility of product consumption is much higher than that of platform consumption, attention becomes highly sensitive to platform prices. In such environments, platforms optimally rely solely on advertising revenue and set prices to zero.

4. **Firm and Platform Entry:** The fourth extension, found in Supplemental Appendix K, introduces entry by both firms and platforms subject to upfront costs. I characterize steady-state equilibrium and show that more informative data can induce firm entry and platform exit, while greater platform substitutability can trigger exit by both firms and platforms.

X. Concluding Discussion

This paper proposes a framework for the modern attention economy in which platforms offer free services to attract attention and monetize that attention through targeted advertising. The framework captures the interaction between the attention and product markets, platforms' dynamic investment incentives, and welfare tradeoffs between paid products and free platform services. These features are salient in ongoing policy debates but rarely analyzed together in formal models.

The paper provides a tool for investigating mechanisms that arise from these features. The results suggest a common theme: The importance of accounting for

cross-market spillovers and dynamic effects when analyzing the attention economy. First, I study data and interoperability policies and find that their effects can be counterintuitive and change sign over time, trading off product consumption against platform quality. In practice, interoperability policies are often aimed at large platforms where strategic effects may be important; evidence from the duopoly variant suggests that similar patterns can arise in such settings.

Second, I investigate welfare in general equilibrium and find that investment and advertising can be too high or too low. With Cobb-Douglas utility, I derive a simple sufficient statistic—based on a few interpretable quantities such as the utility weight, ad revenue, income, elasticity of attention, and product markup—that determines the sign of the investment distortion.

Third, I extend the model to capture data asymmetry between two groups of platforms and highlight a channel through which product-market spillovers may help reconcile the rise of data-rich digital platforms with the decline in advertising's share of GDP in the period 2000–2018.

In addition to these results, the framework can flexibly accommodate several salient features of digital markets. I consider extensions with network effects, platform heterogeneity, and endogenous entry, and examine when zero pricing for platform services can arise endogenously.

There are many promising directions for future work. Endogenizing ad display rates in a duopoly setting would allow a more complete analysis of strategic interaction.³⁹ Although CES preferences preclude personalized pricing, a topic of growing policy attention, the model can be extended to study product steering, a related phenomenon. To study robustness of the findings to other modes of discovery, the framework could also be extended to give consumers an outside option to purchase products without first viewing ads. A sharper characterization of when more informative data reduces ad revenue would further enrich the analysis of data policies.⁴⁰ Finally, modifying the investment technology to accommodate long-run growth could connect the framework to policy discussions around innovation.

Much remains to be explored. I hope the framework offered here can serve as a useful step toward a deeper understanding of the attention economy.

APPENDIX A

A1. *Intuition for Proposition 1*

Consumer i 's CES preferences imply a demand for product $j \in \Omega_{it}$ of

$$(27) \quad c_{ijt} = \frac{I v_{ij}}{\int_{\Omega_{it}} v_{iz} p_{zt}^{1-\sigma} dz} p_{jt}^{-\sigma}.$$

³⁹ See Subsection VIID, which outlines an approach for the case of duopolistic competition.

⁴⁰ Toward this end, methods from Bergemann et al. (2022); Ganuza and Penalva (2010), and the literature on information in auctions may be useful.

Given the demand in (27), firm j 's flow profit from selling to consumer i is

$$(28) \quad \frac{Iv_{ij}}{\int_{\Omega_{it}} v_{iz} p_{zt}^{1-\sigma} dz} p_{jt}^{-\sigma} (p_{jt} - 1).$$

Since v_{ij} and Ω_{it} appear in a term that scales demand by the same factor for any given price, they are irrelevant to the firm's pricing decision. Firm j 's optimal price is therefore a constant markup irrespective of its information about the consumer's preferences.

A2. Equilibrium Properties along the Transition Path

Theorem 1 characterizes the equilibrium in steady state. Along the transition path, the equilibrium has the following properties:

- Ad display rates and attention are as in parts 1 and 6 of Theorem 1.
- The mass of firms in Ω_{it} is

$$(29) \quad M_t = \frac{A}{\lambda_f} - \left(\frac{A}{\lambda_f} - M_0 \right) e^{-\lambda_f t}.$$

- The c.d.f. H_t^c of expected values in Ω_{it}^c satisfies

$$(30) \quad \int_{H_0^c(\hat{v})}^{H_t^c(\hat{v})} \frac{1}{JG(\hat{v}) - Mu^N - (J - M)u} du = \frac{\ln \left[\frac{M - M_0 + (J - M)e^{\lambda_f t}}{J - M_0} \right]}{J - M}$$

at each $\hat{v} \in [0, \infty)$. Given H_t^c , H_t is found from (4).

- Firm j 's bid in an auction for a consumer with expected value $\hat{v}$ is

$$(31) \quad B_t(\hat{v}) = \int_0^{\hat{v}} \int_t^\infty \pi_{\mathbb{J}s} e^{-\int_t^s [\rho + \lambda_f + \lambda_{ez}(y)] dz} ds dy, \quad \hat{v} \in [0, \infty).$$

- The average ad price in an auction is

$$(32) \quad \pi_{\mathbb{K}t} = \int_0^\infty \int_t^\infty \pi_{\mathbb{J}s} e^{-\int_t^s [\rho + \lambda_f + \lambda_{ez}(\hat{v})] dz} ds [1 - O_t(\hat{v})] d\hat{v},$$

where $O_t = (H_t^c)^N + N(H_t^c)^{N-1}(1 - H_t^c)$ denotes the c.d.f. of the second-highest expected value among firms in an auction.

- Platform investment ℓ_{kt} and quality q_{kt} solve the boundary value problem

$$(33) \quad \dot{\ell}_{kt} = \frac{\rho + \delta}{1 - \varphi} \ell_{kt} - \frac{\varphi}{1 - \varphi} \frac{\pi_{\mathbb{K}t} A(\epsilon - 1)}{K q_{kt}} \ell_{kt}^\varphi,$$

$$\dot{q}_{kt} = \ell_{kt}^\varphi - \delta q_{kt},$$

$$\lim_{t \rightarrow \infty} \ell_{kt} = \ell_{\mathbb{K}},$$

$$q_{k0} = q_0,$$

where $\ell_{\mathbb{K}}$ is steady-state investment from part 5 of Theorem 1. Numerical solutions to (33) are computable in seconds via the shooting method.

A3. Proof of Theorem 1

PROOF:

The proof follows the steps outlined in Section IV. I begin with step 3, since step 1 is standard and step 2 was completed in the main text. Steps 1 and 2 yield parts 1 and 6 of Theorem 1.

Step 3: Calculate: M_t, H_t, H_t^c . Because the ODE (7) is linear, it can be solved directly to yield (29). Taking $t \rightarrow \infty$ yields the steady-state value of M in part 2 of Theorem 1.

To derive H_t and H_t^c , I conjecture (and later verify) that bidding strategies are monotone, so that (6) applies:

$$\frac{d}{dt}(M_t H_t(\cdot)) = A H_t^c(\cdot)^N - \lambda_f M_t H_t(\cdot).$$

Using the accounting identity (4) to express this in terms of H_t^c alone gives

$$\frac{d}{dt}[(J - M_t) H_t^c(\cdot)] = \lambda_f [JG(\cdot) - (J - M_t) H_t^c(\cdot)] - A H_t^c(\cdot)^N.$$

Expanding the left-hand side and substituting $\dot{M}_t = A - \lambda_f M_t$ from (7),

$$\begin{aligned} (J - M_t) \dot{H}_t^c(\cdot) &= \lambda_f [JG(\cdot) - (J - M_t) H_t^c(\cdot)] - A H_t^c(\cdot)^N + (A - \lambda_f M_t) H_t^c(\cdot) \\ &= \lambda_f [JG(\cdot) - M H_t^c(\cdot)^N - (J - M) H_t^c(\cdot)], \end{aligned}$$

where $M = A/\lambda_f$ is the steady-state value of M_t . Setting the right-hand side to zero gives the steady-state c.d.f. H^c in part 3 of the theorem.

Since M on the right-hand side is constant, the ODE is separable and can be solved analytically, yielding (30). As $t \rightarrow \infty$, the right-hand side of (30) diverges, so the integral on the left-hand side must diverge as well. The integrand is bounded except where its denominator vanishes, so $H_t^c(\hat{v})$ must approach a zero of the denominator. That zero is the steady-state value $H^c(\hat{v})$ identified above, so H_t^c (and therefore H_t) converges to its steady-state level.

Step 4: Firms' Bidding Strategies. Recall from Section III that the rate at which a firm outside Ω_{it} enters Ω_{it} , conditional on $\hat{v}_{ij} = \hat{v}$, is $\lambda_{et}(\hat{v}) = \lambda_{at} W_t(\hat{v})$, where $\lambda_{at} \equiv N A_t / (J - M_t)$ is the rate at which the firm is invited to bid and $W_t(\hat{v})$ is the probability of winning conditional on being invited. The value functions V_t^{In} and V_t^{Out} satisfy the HJB equations

$$\dot{V}_t^{\text{In}}(\hat{v}) = \rho V_t^{\text{In}}(\hat{v}) - \lambda_f [V_t^{\text{Out}}(\hat{v}) - V_t^{\text{In}}(\hat{v})] - \pi_{\mathbb{J}t} \hat{v}$$

and

$$\dot{V}_t^{\text{Out}}(\hat{v}) = \rho V_t^{\text{Out}}(\hat{v}) - \underbrace{\lambda_{at} W_t(\hat{v})}_{=\lambda_{et}(\hat{v})} \left(V_t^{\text{In}}(\hat{v}) - V_t^{\text{Out}}(\hat{v}) - \mathbb{E} \left[B_t^{(1)} | B_t(\hat{v}) > B_t^{(1)} \right] \right),$$

where $B_t^{(1)}$ is the maximum of the $N - 1$ other bids in that auction. If B_t is increasing, then $W_t = (H_t^c)^{N-1}$ and $B_t^{(1)} \stackrel{d}{=} B_t(\hat{v}^{(1)})$, where $\hat{v}^{(1)} \sim W_t$.

In any equilibrium, because the auction is second price, $B_t = V_t^{\text{In}} - V_t^{\text{Out}}$. Subtracting the two HJB equations gives

$$\dot{B}_t(\hat{v}) = [\rho + \lambda_{at} W_t(\hat{v}) + \lambda_f] B_t(\hat{v}) - \lambda_{at} \int_0^{B_t(\hat{v})} s d\tilde{W}_t(s) - \pi_{\mathbb{J}t} \hat{v},$$

where $\tilde{W}_t$ is the c.d.f. of $B_t^{(1)}$, so $W_t(\hat{v}) = \tilde{W}_t(B_t(\hat{v}))$ by definition. Differentiating in $\hat{v}$ yields

$$\dot{B}_t'(\hat{v}) = [\rho + \lambda_{at} W_t(\hat{v}) + \lambda_f] B_t'(\hat{v}) - \pi_{\mathbb{J}t},$$

which has solution

$$B_t'(\hat{v}) = e^{\int_0^t [\rho + \lambda_f + \lambda_{az} W_z(\hat{v})] dz} \left(B_0'(\hat{v}) - \int_0^t \pi_{\mathbb{J}s} e^{-\int_0^s [\rho + \lambda_f + \lambda_{az} W_z(\hat{v})] dz} ds \right).$$

Since B_t is Lipschitz (Lemma 1 in Supplemental Appendix B), the term in parentheses must vanish as $t \rightarrow \infty$; otherwise B_t' would diverge. Therefore,

$$B_t'(\hat{v}) = e^{\int_0^t [\rho + \lambda_f + \lambda_{az} W_z(\hat{v})] dz} \int_t^\infty \pi_{\mathbb{J}s} e^{-\int_0^s [\rho + \lambda_f + \lambda_{az} W_z(\hat{v})] dz} ds,$$

which is positive, so the bidding function is increasing as conjectured. This holds for any path of H_t , confirming that the assumption underlying (6) is valid.

The bidding function (31) then follows by substituting $H_t^c(\hat{v})^{N-1} = W_t(\hat{v})$. A direct calculation yields the average ad price (32) and its steady-state counterpart in part 4 of Theorem 1.

Step 5: Investment and Quality. Given the average ad price and consumers' attention choices, the Hamiltonian for platform k 's problem is

$$\mathcal{H}(t, q_{kt}, \ell_{kt}, \lambda_t) = \pi_{\mathbb{K}t} A \frac{q_{kt}^{\epsilon-1}}{\int_{\mathbb{K}} q_{lt}^{\epsilon-1} dl} - \ell_{kt} + \lambda_t (\ell_{kt}^\varphi - \delta q_{kt}),$$

where the costate λ_t satisfies

$$\rho \lambda_t - \dot{\lambda}_t = \pi_{\mathbb{K}t} A (\epsilon - 1) \frac{q_{kt}^{\epsilon-2}}{\int_{\mathbb{K}} q_{lt}^{\epsilon-1} dl} - \lambda_t \delta.$$

Lemma 2 in Supplemental Appendix B shows that the platform's problem (12) has a unique solution when $\epsilon - 1 < 1/\varphi$, so all platforms employ the same investment strategy in equilibrium. The costate equation therefore simplifies to

$$\rho\lambda_t - \dot{\lambda}_t = \frac{\pi_{\mathbb{K}t}A(\epsilon - 1)}{Kq_{kt}} - \lambda_t\delta.$$

The first-order condition for the Hamiltonian gives

$$\lambda_t = \frac{1}{\varphi} \ell_{kt}^{1-\varphi},$$

and differentiating in t yields

$$\dot{\lambda}_t = \frac{1-\varphi}{\varphi} \ell_{kt}^{-\varphi} \dot{\ell}_{kt}.$$

Substituting into the costate equation gives

$$(34) \quad \dot{\ell}_{kt} = \frac{\rho + \delta}{1-\varphi} \ell_{kt} - \frac{\varphi}{1-\varphi} \frac{\pi_{\mathbb{K}t}A(\epsilon - 1)}{Kq_{kt}} \ell_{kt}^{\varphi},$$

where, as before,

$$(35) \quad \dot{q}_{kt} = \ell_{kt}^{\varphi} - \delta q_{kt}, \quad q_{k0} = q_0.$$

The steady-state values of ℓ_{kt} and q_{kt} follow from (34)–(35), yielding part 5 of the theorem.

It remains to establish existence and uniqueness of an equilibrium trajectory satisfying (34)–(35). Three lemmas, proved in Supplemental Appendix B, deliver this. Lemma 3 shows that a solution exists for any initial conditions. Lemma 4 shows that any such solution either vanishes, diverges, or converges to steady state. Only the last is consistent with equilibrium, so the equilibrium trajectory solves the boundary value problem (33). Lemma 5 establishes that this boundary value problem has a unique solution.

The final step verifies that the Hamiltonian approach is valid—that is, that platform k optimizes by investing according to the solution to (33), provided its rivals do the same. When $\epsilon < 2$, the Hamiltonian is jointly concave in state and control, so the Mangasarian sufficient conditions apply directly. In the more general case $\epsilon - 1 < 1/\varphi$, joint concavity may fail. I instead apply the calculus of variations: The Gateaux derivative of platform k 's objective with respect to the control vanishes in all directions when investment satisfies (33), and the second-order condition holds because the objective is concave in the control (Lemma 2). Details are omitted for brevity.

Finally, part 7 of the theorem, which records steady-state product and platform consumption aggregates, follows immediately from the earlier parts. ■

A4. Data Shock: Garbling

Figure 6 shows how $\tilde{v} \sim U[0, 1]$ is garbled into $\hat{v} \sim U[0.2, 0.8]$.

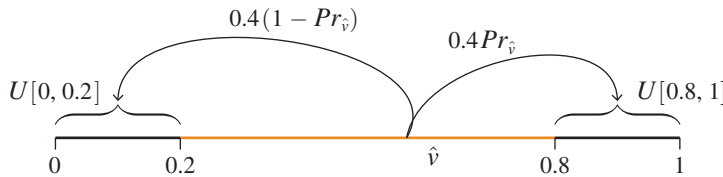


FIGURE 6. JOINT DISTRIBUTION OF PRE- AND POST-SHOCK EXPECTATIONS

Notes: Let $\hat{v}$ be any point in the orange region $[0.2, 0.8]$. Following the shock, $\hat{v}$ stays put, in that $\bar{v} = \hat{v}$ with probability 0.6. Otherwise, with the residual probability 0.4, it jumps to one of the black regions $[0, 0.2] \cup [0.8, 1]$. Conditional on jumping up (down), $\bar{v}$ is distributed uniformly across the upper (lower) black region. The probability $\Pr_{\hat{v}}$ that $\hat{v}$ jumps up is such that the martingale property holds: $\mathbb{E}[\bar{v} | \hat{v}] = \hat{v}$.

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